

NEURAL NETWORKS

Foundations: Neural Networks to Representation Learning

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2026-I



Outline

1 Neural Network Fundamentals

2 Optimization & Regularization



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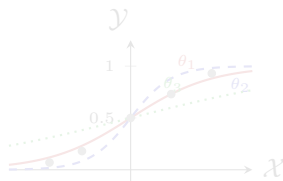
What is a Neural Network?

Definition

A neural network is a parametric function $f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y}$ composed of simple, interconnected computational units (neurons).

↳ ~ 1948

- **Neuron:** weighted sum + nonlinear activation
- **Layer:** neurons operating in parallel
- **Depth:** key enabler of hierarchical features
- **Training:** adjust θ so $f_{\theta}(x) \approx y$



Different $\theta \Rightarrow$ different f_{θ} .

Training finds the best fit.



What is a Neural Network?

⊖ ~~Not~~ convex

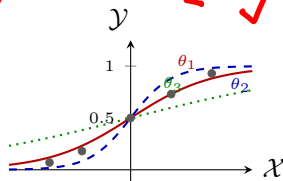
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⊖ convex
 $w_1 + w_2 = 1$



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$$f(x_1, x_2) = w_1 \cdot x_1 + w_2 \cdot x_2$$



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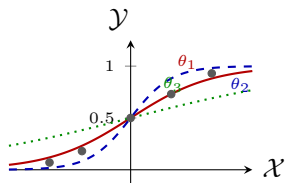
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Parallel



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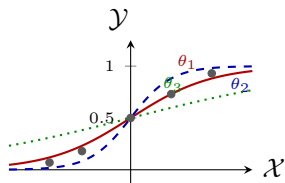


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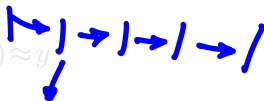
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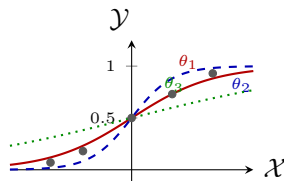


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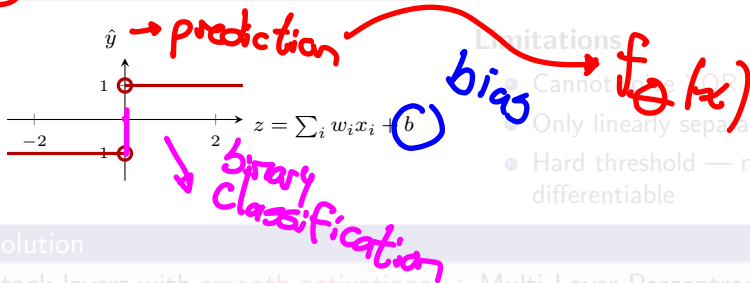
$w_1, w_2 \rightarrow 47$
 $\theta_3 \rightarrow b_i$
 Points



The Perceptron

Rosenblatt (1958)

$$\hat{y} = \text{sign}(\sum_i w_i x_i + b)$$



Strengths

- Binary classification
- Convergence guaranteed (linearly separable data)

Limitations

- Cannot solve XOR
- Only linearly separable
- Hard threshold — not differentiable

Solution

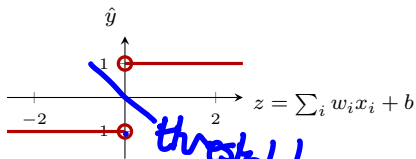
Stack layers with smooth activations ⇒ Multi-Layer Perceptron



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threshold
AND-OR

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Stack layers with smooth activations $\Rightarrow$ Multi-Layer Perceptron

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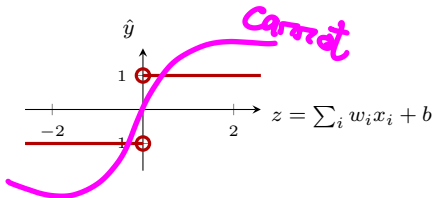
Linear



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Multi-Layer Perceptron (MLP) — Architecture

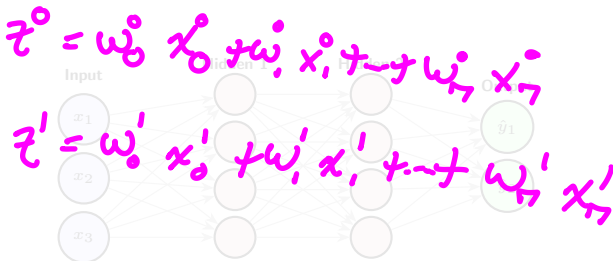
Architecture

Input layer → Hidden layer(s) → Output layer

For each layer l :

$$\mathbf{z}^{(l)} = \mathbf{W}^{(l)} \mathbf{a}^{(l-1)} + \mathbf{b}^{(l)}, \quad \mathbf{a}^{(l)} = \sigma(\mathbf{z}^{(l)})$$

at least
3 layers



• Universal Approximation (Hornik, 1991): one hidden layer with enough neurons approximates any continuous function
 • Depth is more parameter-efficient than extreme width



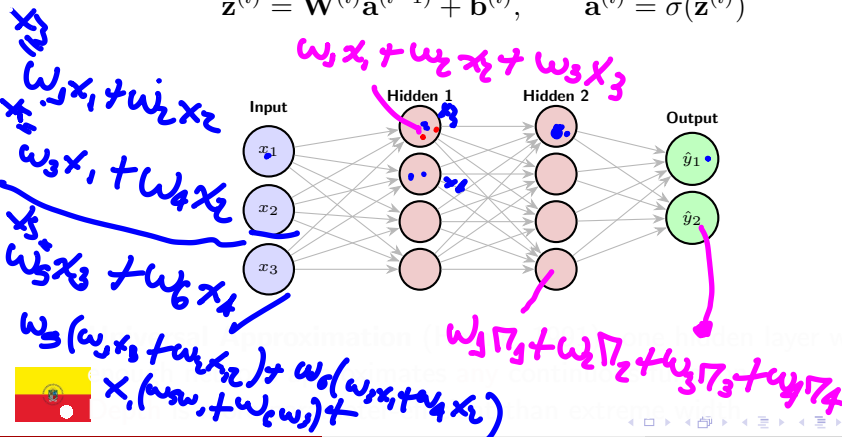
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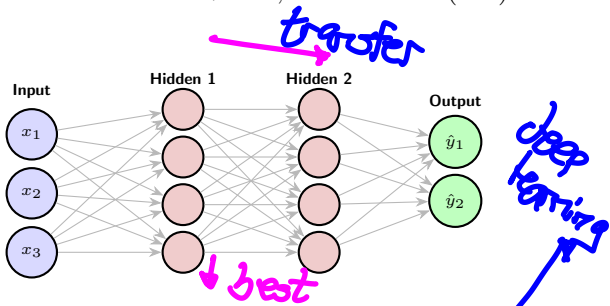
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Multi-Layer Perceptron (MLP) — Output & Softmax

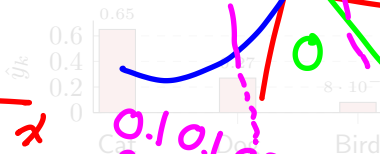
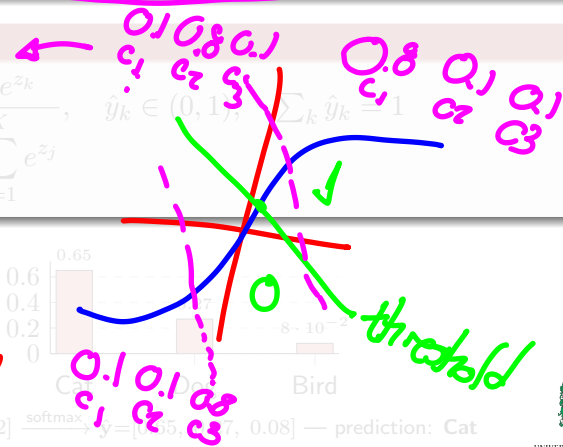
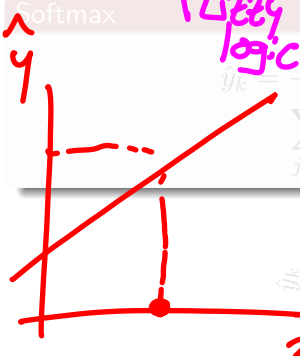
Output layer choices

- **Linear:** regression
- **Sigmoid:** binary classification
- **Softmax:** multi-class → probability distribution over K classes

Softmax

Fuzzy logic

$$\hat{y}_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}, \quad \hat{y}_k \in (0, 1), \quad \sum_k \hat{y}_k = 1$$



$z = [2.1, 0.3, -1.2] \xrightarrow{\text{softmax}} y = [0.65, 0.17, 0.08] \text{ — prediction: Cat}$



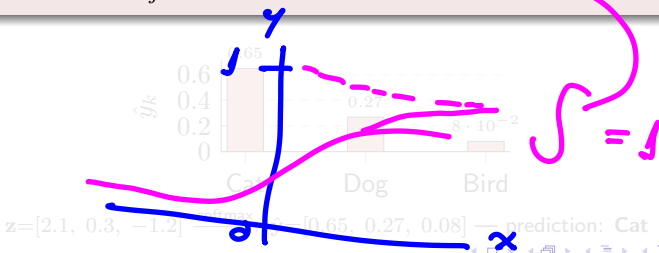
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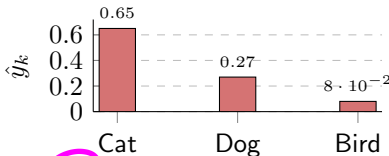
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Activation Functions

Sigmoid:

$$\sigma(x) = \frac{1}{1 + e^{-x}} \in (0, 1)$$

Tanh:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \in (-1, 1)$$

ReLU: $\max(0, x)$ — **default today**

GELU: $x \Phi(x)$ — used in Transformers

→ factor → neuron output

Why ReLU dominates

- No saturation for positive inputs
- Sparse activation (many zeros)
- Computationally trivial
- Mitigates vanishing gradient

$(w_1 x_1 + w_2 x_2) \cdot \text{af}$



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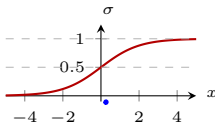
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Activation Functions — Visual Overview

Sigmoid

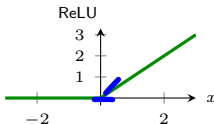
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Output layer

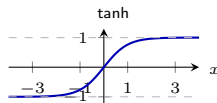
ReLU

$$\max(0, x)$$



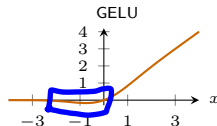
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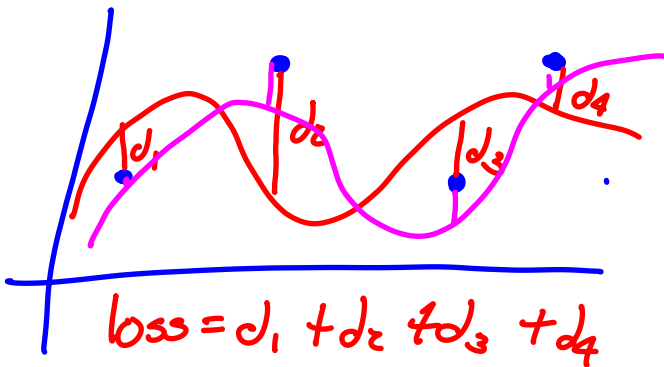
$$x \cdot \Phi(x)$$



Loss Functions

The Objective of Training

Minimise $\mathcal{L}(\theta)$ — a measure of disagreement between predictions and targets over the training set.



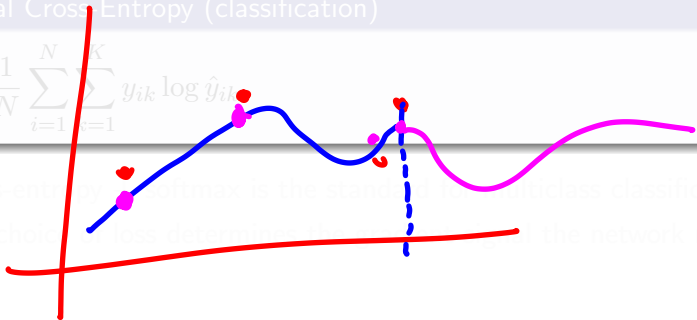
Loss Functions

Mean Squared Error (regression)

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N \|\hat{y}_i - y_i\|^2$$

Categorical Cross Entropy (classification)

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{ik} \log \hat{y}_{ik}$$



- Cross-entropy with softmax is the standard loss for multiclass classification
- The choice of loss determines the metric that the network receives



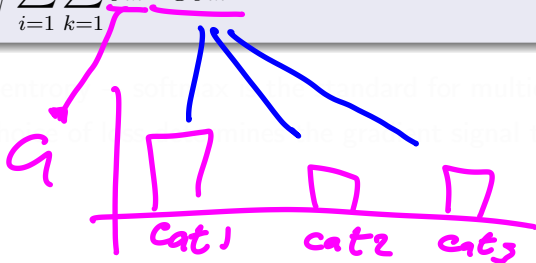
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- The choice of loss determines the gradient signal the network receives



Backpropagation

Core Idea

Apply the chain rule repeatedly to compute $\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(l)}}$ for every layer, from output back to input.

Two-pass algorithm:

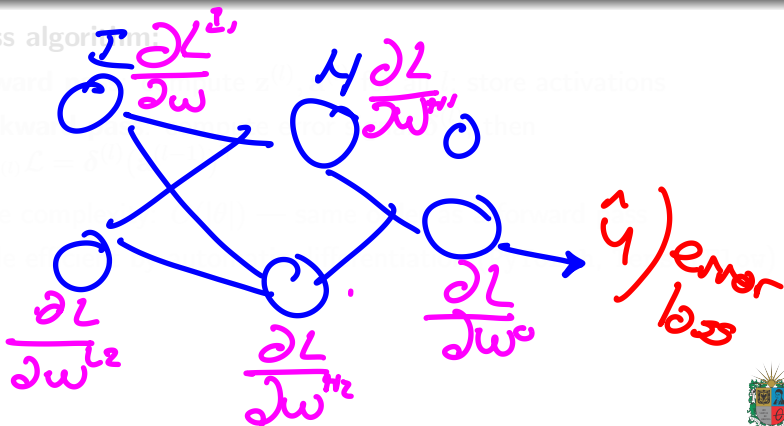
• Forward pass: compute all $\mathbf{z}^{(l)}$ and store activations

• Backward pass: compute all $\delta^{(l)}$ when

$\nabla_{\mathbf{W}^{(l)}} \mathcal{L} = \delta^{(l)} \mathbf{W}^{(l)}$

Time complexity is $O(N^2)$ for fully connected layers

Made possible by GPUs



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Apply the **chain rule** repeatedly to compute $\partial \mathcal{L} / \partial \mathbf{W}^{(l)}$ for every layer, from output back to input.

Two-pass algorithm:

① **Forward pass:** compute $\mathbf{z}^{(l)}$, $\mathbf{a}^{(l)}$ for all l ; store activations

② **Backward pass:** compute error signal $\delta^{(l)}$, then

$$\nabla_{\mathbf{W}^{(l)}} \mathcal{L} = \delta^{(l)} (\mathbf{a}^{(l-1)})^\top$$

• Time complexity: $O(|\theta|)$ — same order as a forward pass

• Made efficient by automatic differentiation (PyTorch, TensorFlow)



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The Training Loop

- 1 **Initialise** weights θ_0 (random)
- 2 Split data: train / validation / test
- 3 For each **epoch**:
 - Sample a mini-batch $\{(x_i, y_i)\}_{i=1}^B$
 - Forward pass $\rightarrow$ compute $\mathcal{L}$
 - Backward pass $\rightarrow$ compute $\nabla \mathcal{L}$
 - Update $\theta \leftarrow \theta - \eta \nabla \mathcal{L}$
- 4 Monitor **validation loss**; stop when it increases (early stopping)

Golden Rule

Never tune hyperparameters using the **test set**. The test set measures final generalisation only.



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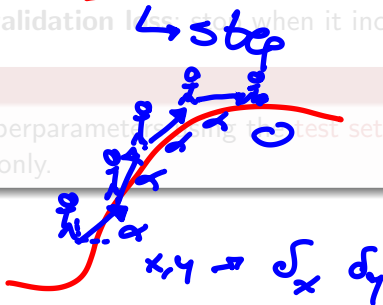
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over all train data

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overfitting



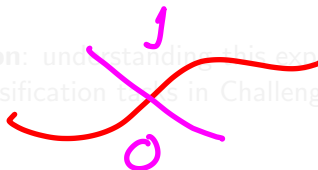
Practical Example: MLP on Two-Moons

- **Problem:** binary classification on non-linear boundary
- **Architecture:** $2 \rightarrow 32 \rightarrow 32 \rightarrow 1$, ReLU, sigmoid output
- **Experiment:** vary number of hidden layers and width
- **Observation:** deep narrow network generalises better than wide shallow one
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Gradient Descent

Update Rule

$$\theta \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}(\theta)$$

Full-batch GD

$$\theta_0 = w_1, w_2, w_3, \dots, w_n$$

Deterministic

$$\theta_1 = w_1, w_2, w_3, \dots, w_n$$

Mini-batch SGD

- Noisy gradient estimates $\rightarrow$ random
- Constant step size $\rightarrow$ per step
- Noise helps escape local minima $\rightarrow \nabla_{\theta}$
- Standard in practice

- Batch size B : typical values 32–512; affects convergence and memory
- Learning rate α is the most critical hyperparameter

$$\nabla_{\theta} \approx \theta$$



Gradient Descent

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hyperparameter optimization

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$\bar{\nabla} \neq 0$

32-64-128-256-512

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0.0003
0.0001

0.003
0.001

0.03
0.01

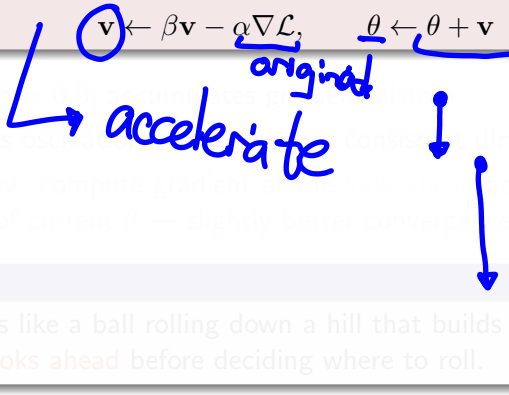
0.3
0.1



Momentum and Nesterov *~ 80's*

SGD with Momentum

$$\mathbf{v} \leftarrow \beta \mathbf{v} - \alpha \nabla \mathcal{L}, \quad \theta \leftarrow \theta + \mathbf{v}$$



Intuition

Momentum is like a ball rolling down a hill that builds *inertia*. Nesterov is a ball that *looks ahead* before deciding where to roll.



Momentum and Nesterov

SGD with Momentum

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- Typical $\beta = 0.9$; accumulates gradient history
- Dampens oscillations; accelerates in consistent directions
- **Nesterov**: compute gradient at the **look-ahead** position $\theta + \beta \mathbf{v}$ instead of current θ — slightly better convergence

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Momentum is like a ball rolling down a hill that builds **inertia**. Nesterov is a ball that **looks ahead** before deciding where to roll.



Momentum and Nesterov

SGD with Momentum

$$\mathbf{v} \leftarrow \beta \mathbf{v} - \alpha \nabla \mathcal{L}, \quad \theta \leftarrow \theta + \mathbf{v}$$

- Typical $\beta = 0.9$; accumulates gradient **history**
- Dampens oscillations; accelerates in consistent directions
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↓
snowball



Adam Optimizer $\rightarrow$ XGBoost

Adaptive Moment Estimation (Kingma & Ba, 2015)

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \underline{g}_t$$

$\swarrow$ *weighted avg.*

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

$$\theta \leftarrow \theta - \underline{\alpha} \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon)$$

$$\hat{m}_t = m_t / (1 - \beta_1^t)$$

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$\swarrow$ *scaling*

$\swarrow$ *momentum*

$\swarrow$ *adaptive*

Convex Combination
 $\alpha + (1 - \alpha)$



Adam Optimizer

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- Defaults: $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$, $\alpha = 10^{-3}$
0.001
- Per-parameter adaptive learning rates
- **AdamW**: Adam + decoupled weight decay — preferred in current practice
- Adam dominates in deep learning; SGD+Momentum preferred in CV research



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Learning Rate Scheduling

- **Step decay:** multiply LR by factor γ every k epochs

- **Cosine annealing:** $\alpha_t = \alpha_{\min} + \frac{1}{2}(\alpha_{\max} - \alpha_{\min})(1 + \cos(\pi t/T))$

- Linear warmup ramp from near-zero for first w steps, then decay

- ReduceLROnPlateau: halve LR when validation loss stops improving

Warmup and Decay

Modern practice: **linear warmup + cosine decay**. Used in GPT, BERT, and GPT. Especially important with large batch sizes.

Converge fast

Optimal



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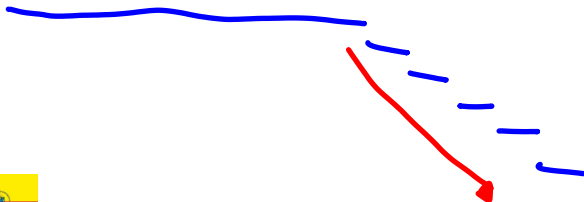


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Google



Bias-Variance Trade-Off

Underfitting (high bias)

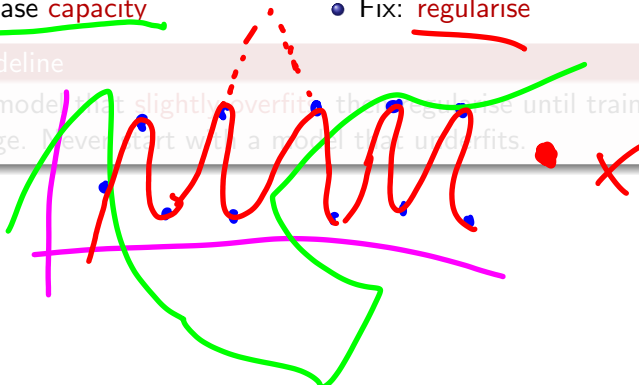
- Model too simple
- High training error
- Cannot capture data structure
- Fix: increase capacity

Overfitting (high variance)

- Model too complex
- Low training, high val error
- Memorises noise
- Fix: regularise

Practical Guideline

Start with a model that slightly overfits, then regularise until train and val errors converge. Never start with a model that underfits.



Bias-Variance Trade-Off

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α -medium $\Leftarrow$ Control



L1 and L2 Regularization

L1 and L2 Regularization → training time

Augmented Objective

$$\mathcal{L}_{\text{reg}}(\theta) = \mathcal{L}(\theta) + \Omega(\theta)$$

$\mathcal{L} \rightarrow \Omega(\theta)$

L2 Regularization (Weight Decay / Ridge)

$$\Omega(\theta) = \frac{\lambda}{2} \|\mathbf{W}\|_F^2 \quad \text{drives weights small but rarely zero}$$

$\mathcal{L} \rightarrow -\Omega(\theta)$

L1 Regularization (Lasso)

$$\Omega(\theta) = \lambda \|\mathbf{W}\|_1 \quad \text{promotes sparse weights (many exact zeros)}$$



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- Elastic Net: combines both — used in linear models and sparse NNs



L1 and L2 Regularization

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Penish

stable control over-fitting

L2 Regularization (Weight Decay / Ridge)

$$\Omega(\theta) = \frac{\lambda}{2} \|\mathbf{W}\|_F^2$$

drives weights **small** but rarely zero

feature selection

L1 Regularization (Lasso)

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promotes **sparse** weights (many exact zeros)

*$\lambda \rightarrow$ hyper parameter
high $\rightarrow$ bias
low $\rightarrow$ variance*

edge noise

hyp opt.

$\hookrightarrow$ soft-learning $\rightarrow$ Bayes



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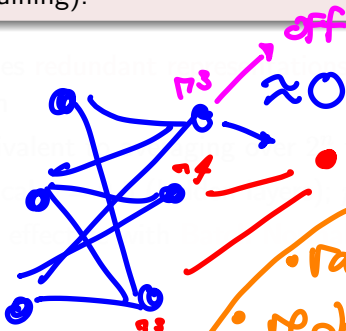
Dropout

→ regularization

 $\frac{1}{c}$

Definition (Srivastava et al., 2014)

During training, independently zero out each activation with probability p .
 At inference, multiply activations by $(1 - p)$ (*inverted dropout* scales during training).



$$w_1 n_4 + w_2 n_3 + w_3 n_3$$

- random → off n neurons
- reduce overfitting



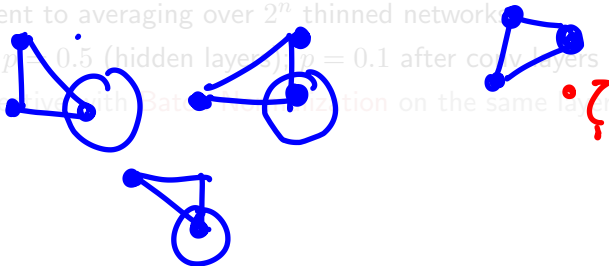
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- Equivalent to averaging over 2^n thinned networks
- Typical $p = 0.5$ (hidden layers), $p = 0.1$ after conv layers
- Less effective with Batch Normalization on the same layer



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Batch Normalization

Definition (Ioffe & Szegedy, 2015)

Normalise activations within a mini-batch, then scale and shift:

$$\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \varepsilon}}, \quad y_i = \gamma \hat{x}_i + \beta$$

stable + momentum
 ↓
 regularization



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7 B
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- Reduces sensitivity to weight initialisation

- At inference: use running mean/variance accumulated during training
- Alternative: Layer Norm (Transformers), Group Norm (small batches)



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GLM



Practical Example: Regularizing a Deep MLP

- **Problem:** overfit a 5-layer MLP on a small dataset
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Outline

- 1 Neural Network Fundamentals
- 2 Optimization & Regularization



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/machine-learning>

